

VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report: Use Telnet and SSH

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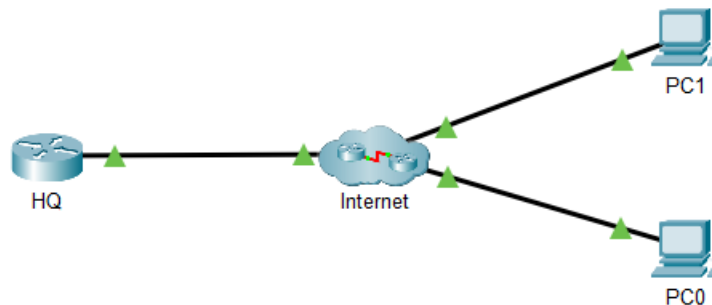
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Introduction

This lab demonstrates how to remotely manage a network device using Telnet and SSH. Telnet is an older, unencrypted protocol, while SSH provides secure, encrypted administrative access to network devices. Students will verify IP connectivity, attempt a Telnet session, observe why it fails, and then successfully connect to the HQ router using SSH.

Addressing Table

Device / Interface	IP Address	Subnet Mask
HQ – G0/0/1	64.100.1.1	255.255.255.0
PC0 – NIC	DHCP	DHCP
PC1 – NIC	DHCP	DHCP



Objectives

- Verify network connectivity.
- Test Telnet remote access.
- Establish a secure SSH connection.
- Understand the differences between Telnet and SSH.

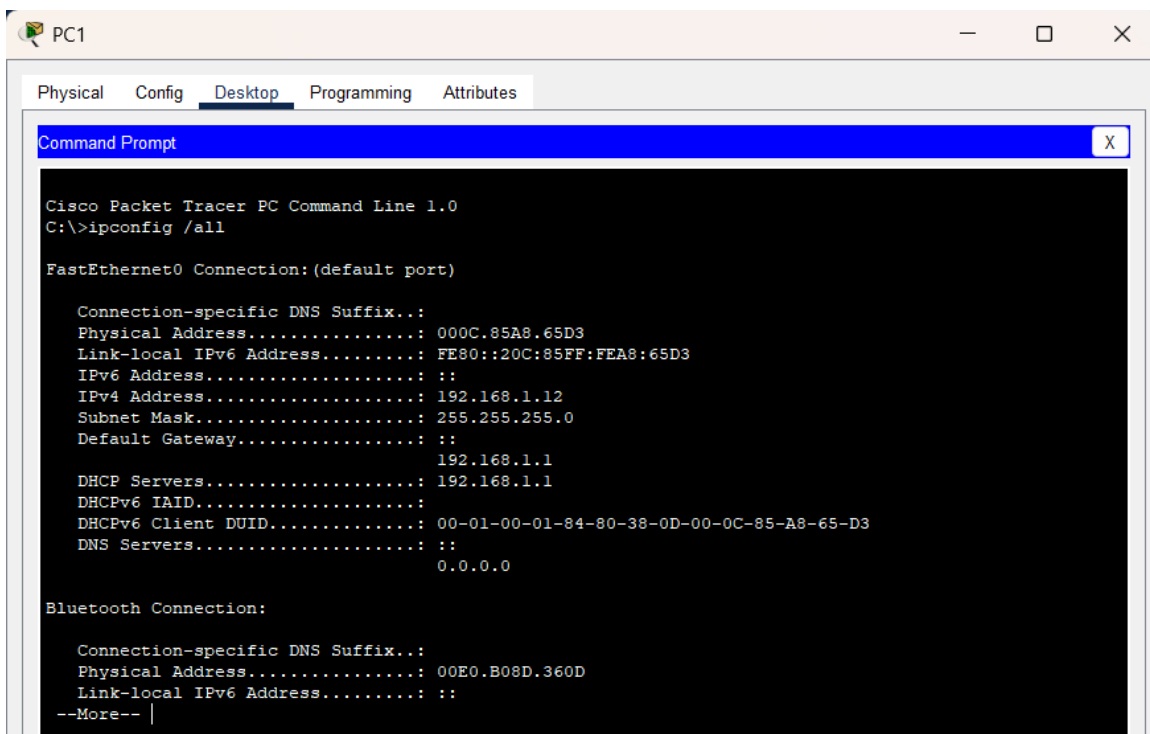
Part 1: Verify Connectivity

Step 1: Verify IP address on a PC

From a PC, click **Desktop**. Click **Command Prompt**.



Command used to verify DHCP IP address: ipconfig

A screenshot of a PC window titled "PC1" with tabs for Physical, Config, Desktop, Programming, and Attributes. The "Desktop" tab is active, showing a "Command Prompt" window. The command prompt displays the output of the "ipconfig /all" command, showing network configuration details for both FastEthernet0 and Bluetooth connections. The FastEthernet0 section shows an IPv4 address of 192.168.1.12 and a default gateway of 192.168.1.1. The Bluetooth section shows a physical address of 00E0.B08D.360D.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Physical Address...                : 000C.85A8.65D3
    Link-local IPv6 Address...         : FE80::20C:85FF:FEA8:65D3
    IPv6 Address...                   : ::
    IPv4 Address...                   : 192.168.1.12
    Subnet Mask...                   : 255.255.255.0
    Default Gateway...                : ::
                                         192.168.1.1
    DHCP Servers...                  : 192.168.1.1
    DHCPv6 IAID...                   : 
    DHCPv6 Client DUID...            : 00-01-00-01-84-80-38-0D-00-0C-85-A8-65-D3
    DNS Servers...                   : ::
                                         0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address...                : 00E0.B08D.360D
    Link-local IPv6 Address...         : ::
--More-- |
```

PC0 ipconfig

Question:

What command did you use to verify the IP address from DHCP?

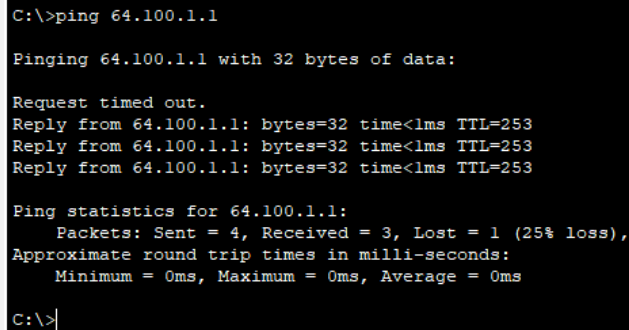
Answer:

ipconfig

Step 2: Verify connectivity to HQ

Use the ping command to test reachability to the HQ router:

```
ping 64.100.1.1
```



```
C:\>ping 64.100.1.1

Pinging 64.100.1.1 with 32 bytes of data:

Request timed out.
Reply from 64.100.1.1: bytes=32 time<1ms TTL=253
Reply from 64.100.1.1: bytes=32 time<1ms TTL=253
Reply from 64.100.1.1: bytes=32 time<1ms TTL=253

Ping statistics for 64.100.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

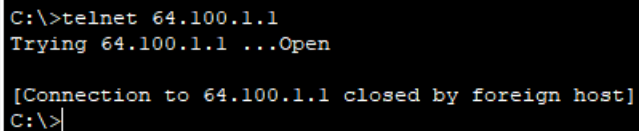
Ping HQ

Part 2: Access a Remote Device

Step 1: Telnet to HQ

At the PC command prompt, enter:

```
telnet 64.100.1.1
```



```
C:\>telnet 64.100.1.1
Trying 64.100.1.1 ...Open

[Connection to 64.100.1.1 closed by foreign host]
C:\>|
```

Telnet Attempt

Question:

Were you successful? What was the output?

Answer:

No, the Telnet connection was not successful.

The router is configured to reject insecure Telnet access, so the output typically shows:

- “Password required, but none set” or
- “Connection refused”

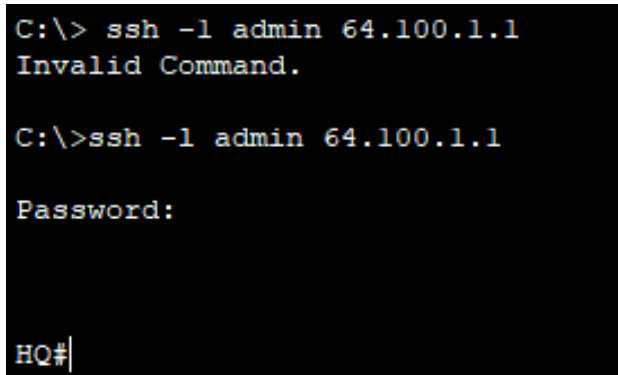
Step 2: SSH to HQ

SSH is configured on the router, so use:

Command: `ssh -l admin 64.100.1.1`

Password: class

Prompt after successful SSH login: HQ#



```
C:\> ssh -l admin 64.100.1.1
Invalid Command.

C:\>ssh -l admin 64.100.1.1

Password:

HQ#
```

successful SSH session.

Question:

What is the prompt after accessing the router successfully via SSH?

Answer:

After logging in with SSH, the router prompt appears as:

HQ#

This indicates successful login to **privileged EXEC mode**.

Observations & Conclusion

This lab highlights the importance of secure remote access. Telnet is rejected because it sends data in plaintext, making it insecure. SSH provides encrypted communications, ensuring safe access to network devices. Verifying IP addressing and connectivity is essential before attempting remote management.